

1

OSPF动态网络路由配置

冯巾松

fengjinsong@tongji.edu.cn

OSPF协议概述

- ➡ OSPF是Open Shortest Path First(开放最短路径优先协议)的简称，管理距离为110；
- ➡ OSPF路由协议是一种典型的链路状态（Link state）的路由协议。使用接口带宽来衡量路径开销，带宽更优（开销越小）的路径被优选
- ➡ OSPF目前有2个版本
 - ✓ V2：适用与IPv4的环境
 - ✓ V3：适用与IPv6的环境
- ➡ OSPF采用分级管理机制

技术原理

3

➡ 作为一种链路状态的路由协议，OSPF将链路状态组播数据LSA（Link State Advertisement）传递给在某一区域内的所有路由器，这一点与距离矢量路由协议不同。运行距离矢量路由协议的路由器是将部分或全部的路由表传递给与其相邻的路由器。

OSPF协议特点

- 触发更新（路由表学习完成后，若网络不发生变化，则不再学习）
- 路径选择的依据是开销，不是距离

OSPF的配置

(1) 开启OSPF进程：在全局模式下使用
router ospf 进程号

- ➡ 设备上可以运行多个OSPF进程，因此要加个进程号用于区分。不同进程间相当于不同路由协议，不会交互路由信息
- ➡ 进程号只具有本地意义，本地不冲突就可以，不会在邻居间传递

OSPF的配置

(2) 发布路由信息

格式为：**network 网络位 反掩码 区域号**

- 使用network命令发布路由信息，和RIP配置中的作用一样，用于发布路由和建立邻居关系。
- 网络位和反掩码合并起来，表示一个IP地址的范围，如果接口IP地址在这个范围内，则OSPF将这个接口IP地址所在网段发布给邻居，并且尝试在这条链路上和对方建立邻居关系。
- 区域号表示将匹配这些网段的接口划分到该区域内。OSPF为了在大型网络中提高效率，将网络按照区域分隔开。

反掩码

7

- ➡ **正掩码**用来区分主机位和网络位，使用1表示对应的网络位，0表示主机位

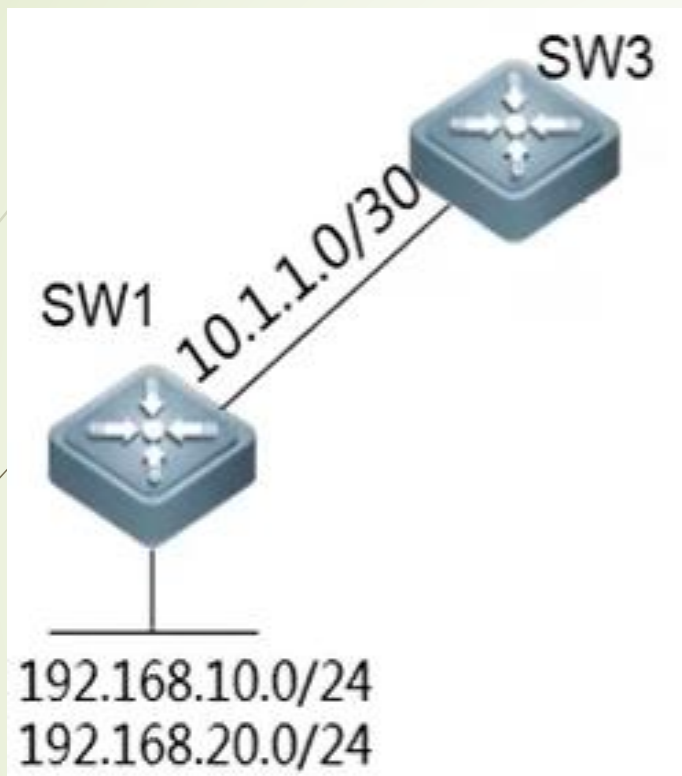
192.168.10.2	→	11000000	10101000	00001010	00000010
255.255.255.0	→	11111111	11111111	11111111	00000000

- ➡ **反掩码**用来表示一段IP地址和范围，掩码中的1对应不用匹配的位置，0对应必须匹配位置

192.168.10.2	→	11000000	10101000	00001010	00000010
0.0.0.255	→	00000000	00000000	00000000	11111111

OSPF的配置示例

8



SW1 (config)#router ospf 1
//1, 开启ospf进程, 进程号是1

反掩码前24位是0, 表示严格匹配; 后8位为1表示不用匹配。因此它们表示192.168.10.0 – 192.168.10.255这段IP范围

SW1 (config-router)#network 192.168.10.0 0.0.0.255 area 0
SW1 (config-router)#network 192.168.20.0 0.0.0.255 area 0
SW1 (config-router)#network 10.1.1.1 0.0.0.0 area 0
//2, 使用network命令发布网段

查看相关信息

- 显示路由表信息

Router#show ip route

- 查看OSPF邻居表

Router#show ip ospf neighbor

- 查看接口OSPF相关信息

Router#show ip ospf interface

- 启动调试OSPF的邻解关系

Router#debug ip ospf adj



Physical Config CLI Attributes

IOS Command Line Interface

```
Router#show ip route
```

```
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route
```

Gateway of last resort is not set

```
10.0.0.0/24 is subnetted, 1 subnets
O    10.60.2.0 [110/65] via 202.120.17.18, 00:25:03, Serial0/0/0
118.0.0.0/24 is subnetted, 1 subnets
C    118.18.4.0 is directly connected, FastEthernet0/1
172.16.0.0/24 is subnetted, 1 subnets
C    172.16.3.0 is directly connected, FastEthernet0/0
O    192.168.1.0/24 [110/65] via 202.120.17.18, 00:25:03, Serial0/0/0
C    202.120.17.0/24 is directly connected, Serial0/0/0
```

```
Router#show ip ospf neighbor
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
202.120.17.18	0	FULL/ -	00:00:37	202.120.17.18	Serial0/0/0

```
Router#
```

```
Router#show ip ospf interface
```

```
FastEthernet0/1 is up, line protocol is up
  Internet address is 118.18.4.254/24, Area 0
  Process ID 1, Router ID 202.120.17.29, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 202.120.17.29, Interface address 118.18.4.254
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 1/1, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
FastEthernet0/0 is up, line protocol is up
  Internet address is 172.16.3.254/24, Area 0
  Process ID 1, Router ID 202.120.17.29, Network Type BROADCAST, Cost: 1
  Transmit Delay is 1 sec, State DR, Priority 1
  Designated Router (ID) 202.120.17.29, Interface address 172.16.3.254
  No backup designated router on this network
  Timer intervals configured, Hello 10, Dead 40, Wait 40, Retransmit 5
    Hello due in 00:00:08
  Index 2/2, flood queue length 0
  Next 0x0(0)/0x0(0)
  Last flood scan length is 1, maximum is 1
  Last flood scan time is 0 msec, maximum is 0 msec
  Neighbor Count is 0, Adjacent neighbor count is 0
  Suppress hello for 0 neighbor(s)
Serial0/0/0 is up, line protocol is up

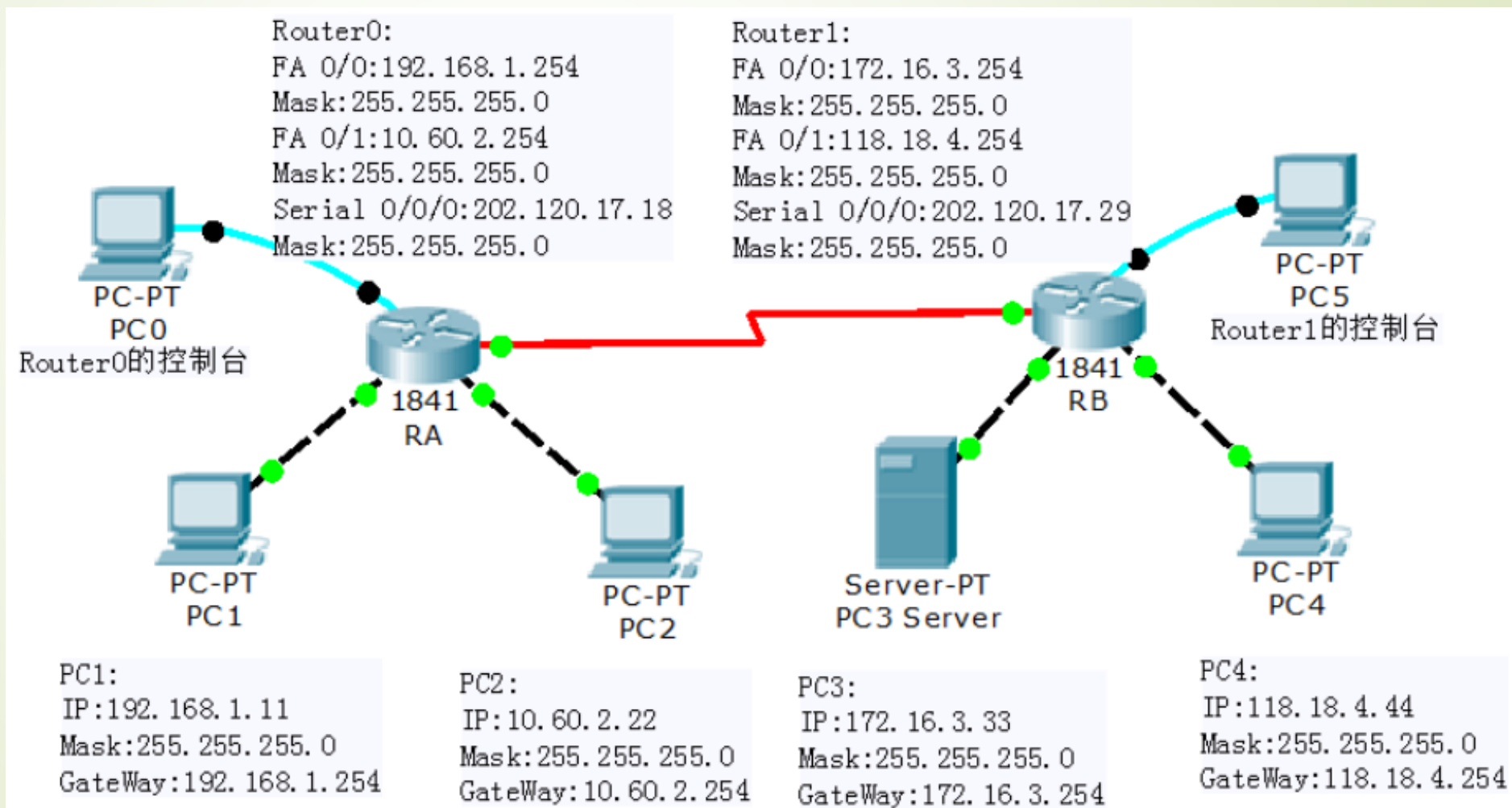
Router#
```

单区域OSPF实验步骤

- ➡ 1 首先规划网络地址及 拓扑图；
- ➡ 2 配置PC机、服务器及路由器IP地址；
- ➡ 3 第一次检查pc间能否相互ping通；
- ➡ 4 在RA （或RB） 上配置OSPF；
- ➡ 5 第二次检查pc间能否相互ping通
- ➡ 6 在RB （或RA） 上配置OSPF；
- ➡ 7 第三次验证主机之间的互通性；
- ➡ 8 查看路由器上的路由信息

实验步骤1

网络拓扑及地址规划



实验步骤2

14

➡ (1) 配置路由器的各端口地址(以RA为例)

```
Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.254 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface FastEthernet0/1
Router(config-if)#ip address 10.60.2.254 255.255.255.0
Router(config-if)#no shutdown
Router(config-if)#exit
Router(config)#interface Serial0/0/0
Router(config-if)#ip address 202.120.17.254 255.255.255.0
Router(config-if)#clock rate 56000
Router(config-if)# no shutdown
```


实验过程4

15

➤ (3) 配置RA的OSPF路由表

RA(config)#router ospf 1 //启用ospf协议并指定AS号

RA(config-router)#network 192.168.1.0 0.0.0.255 area 0

//设定公告网段，指定反掩码，并指定所在area

RA(config-router)#network 10.60.2.0 0.0.0.255 area 0

RA(config-router)#network 202.120.17.0 0.0.0.255 area 0

➤ (4) 配置RB的OSPF路由表

router ospf 1

network 172.16.3.0 0.0.0.255 area 0

network 118.18.4.0 0.0.0.255 area 0

network 202.120.17.0 0.0.0.255 area 0

实验过程8

16

RA

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 1 subnets
C       10.60.2.0 is directly connected, FastEthernet0/1
  118.0.0.0/24 is subnetted, 1 subnets
O       118.18.4.0 [110/65] via 202.120.17.29, 00:01:23, Serial0/0/0
  172.16.0.0/24 is subnetted, 1 subnets
O       172.16.3.0 [110/65] via 202.120.17.29, 00:01:23, Serial0/0/0
C       192.168.1.0/24 is directly connected, FastEthernet0/0
C       202.120.17.0/24 is directly connected, Serial0/0/0

Router#
```

RB

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

  10.0.0.0/24 is subnetted, 1 subnets
O       10.60.2.0 [110/65] via 202.120.17.18, 00:00:05, Serial0/0/0
  118.0.0.0/24 is subnetted, 1 subnets
C       118.18.4.0 is directly connected, FastEthernet0/1
  172.16.0.0/24 is subnetted, 1 subnets
C       172.16.3.0 is directly connected, FastEthernet0/0
O       192.168.1.0/24 [110/65] via 202.120.17.18, 00:00:05, Serial0/0/0
C       202.120.17.0/24 is directly connected, Serial0/0/0

Router#
```

单区域OSPF实验内容

- ➡ 1) 按照13页拓扑图构建网络;
- ➡ 2) 配置各设备的IP和端口。其中PC1的IP为“192.168.1.学号1-2位”; PC2的IP为“10.60.2.学号3-4位”; PC3的IP为“172.16.3.学号4-5位”; PC4的IP为“118.18.4.学号6-7位”, 其他配置与图片信息一致;
- ➡ 3) 检查各PC间的互通性, 并分析原因;
- ➡ 4) 在两台路由器上分别配置OSPF;
- ➡ 5) 再次检查主机之间的互通性, 分析原因
- ➡ 6) 查看两台路由器上的路由和OSPF信息

问题分析讨论

- ➡ 1)在路由器A和B配置OSPF之前，几台计算机互相Ping，测试访问结果。
- ➡ 2)路由器A或B配置OSPF之后，几台计算机互相Ping，测试访问结果。
- ➡ 3)路由器A和B配置OSPF之后，几台计算机互相Ping，测试访问结果。
- ➡ 4)查看路由器RA的路由 (Router#sh ip route) ,对信息进行分析
- ➡ 5)查看路由器RB的邻居 (Router#sh ip ospf neighbor) ,对信息进行分析